Null-Hypothesis Calibration for Long-Baseline Quantum Gravity Interferometry: Separating Relativistic Synchronization Lag from Ξ_{QG} Anomalies

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Abstract

Next-generation long-baseline atom interferometers and optomechanical quantum sensors—including MAGIS-100, AION, and MIGA—aim to probe the interface between quantum mechanics and general relativity by searching for anomalous phase decoherence or gravitational entanglement signatures. A central methodological challenge in these experiments is distinguishing genuine quantum-gravity phenomenology from classical and relativistic metrological noise across kilometer-scale baselines. By applying the Operational Simultaneity Horizon ($\mathcal{L}_{\mathrm{sim}}$) and Spatial Locality Decay Factor ($\mathcal{D}_{\mathrm{loc}}$) formalism established in earlier installments of this series, we derive an exact, parameter-free **Classical Synchronization Null-Hypothesis Visibility** curve (V_{null}). We prove that finite two-way laser phase-locking latency ($\Delta\tau_{\mathrm{sig}}$) combined with technical phase diffusion (Γ_{tot}) induces a deterministic exponential visibility decay across baseline $\Delta\ell$ that closely mimics models of gravitational or spacetime-foam decoherence. To prevent false-positive detections, we introduce the **Two-Tier Quantum Gravity Anomaly Monitor** (Ξ_{QG}) and establish an orthogonal noise-sweep protocol capable of cleanly isolating genuine spacetime-induced anomalies from relativistic synchronization lag.

1 Introduction: The False-Positive Risk in Long-Baseline Quantum Gravity Tests

Experimental quantum gravity has entered an era of precision terrestrial interferometry. Facilities such as the Matter-wave Atomic Gradiometer Interferometric Sensor (MAGIS-100), the Atom Interferometer Observatory and Network (AION), and the Matter-wave laser Interferometric Gravitation Antenna (MIGA) deploy spatially separated atomic ensembles linked by common laser beams across baselines $\Delta\ell$ ranging from 100 m to 1 km and beyond. Alongside tabletop levitated nanoparticle experiments testing gravitationally induced entanglement, these platforms seek empirical signatures of Planck-scale spacetime fluctuations, Diósi–Penrose objective collapse, or anomalous gravitational decoherence.

In all such long-baseline architectures, the observable signature of an anomalous gravitational interaction is a subtle, distance-dependent loss of interference fringe visibility $V_{\mathrm{obs}}(\Delta\ell)$ or excess differential phase variance between spatially separated nodes. However, as established in Papers 9–11 of the *Temporal Coherence & Operational Relativistic Metrology* series, maintaining a shared phase reference across a proper baseline $\Delta\ell$ is strictly bounded by **Coherence-Bounded Operational Locality**.

When experimentalists model laser phase noise using idealized, zero-latency point-locality ($\mathcal{D}_{\text{loc}} \equiv 1$), they systematically underestimate the baseline-dependent phase variance accumulated during the finite round-trip propagation time $\Delta\tau_{\text{sig}}$ of the interrogation and feedback pulses. Left uncalibrated, this deterministic relativistic synchronization lag produces a baseline-dependent exponential visibility roll-off that is mathematically indistinguishable from putative quantum-gravity decoherence models.

2 Mathematical Framework: The Synchronization Null-Hypothesis Baseline

Consider a gradiometric atom interferometer consisting of two local atomic ensembles A and B separated by a proper vertical or horizontal baseline $\Delta\ell$ inside an evacuated optical cavity or waveguide of effective group refractive index N_{eff} (where $N_{\text{eff}} \approx 1$ in ultra-high vacuum).

2.1 Round-Trip Interrogation Delay and Operational Simultaneity Horizon

To execute coherent $\pi/2 - \pi - \pi/2$ Bragg or clock-transition pulse sequences across both ensembles, electromagnetic phase control signals must traverse the baseline $\Delta\ell$. The minimum two-way Einstein–Poincaré synchronization and feedback round-trip time is:

$$\Delta\tau_{\text{sig}} = \frac{2\Delta\ell}{cN_{\text{eff}}}. \tag{1}$$

Let Γ_{tot} (in s^{-1}) denote the total combined classical and technical phase-diffusion rate of the master interrogation laser, seismic/vibrational mirror jitter, and local magnetic/thermal fluctuations, and let $\rho_\tau \in (0, 1]$ denote the target phase-locking threshold parameter. The resulting **Operational Simultaneity Horizon** ($\mathcal{L}_{\text{sim}}$) is:

$$\mathcal{L}_{\text{sim}} = \frac{cN_{\text{eff}}\rho_\tau}{2\Gamma_{\text{tot}}}. \tag{2}$$

Accordingly, the dimensionless **Spatial Locality Decay Factor** ($\mathcal{D}_{\text{loc}}$) governing active phase coordination across baseline $\Delta\ell$ is given by:

$$\mathcal{D}_{\text{loc}}(\Delta\ell, \Gamma_{\text{tot}}) = \exp\left(-\frac{\Delta\ell}{\mathcal{L}_{\text{sim}}}\right) = \exp\left(-\frac{2\Gamma_{\text{tot}}\Delta\ell}{cN_{\text{eff}}\rho_\tau}\right). \tag{3}$$

2.2 Derivation of the Classical Synchronization Null-Hypothesis Visibility

Definition 1 (Classical Synchronization Null-Hypothesis Visibility)**.** *Let $\Phi_\tau(\Omega) \in [0, 1]$ denote the intrinsic local single-node fringe contrast calibrated at zero baseline ($\Delta\ell \to 0$) for Rabi coupling parameter Ω. Under the null hypothesis of purely classical general relativity and standard quantum mechanics (zero quantum-gravity decoherence), the predicted differential interference visibility across baseline $\Delta\ell$ is:*

$$V_{\text{null}}(\Omega, \Delta\ell, \Gamma_{\text{tot}}) \equiv \mathcal{S}_{\text{sim}}(\Omega, \Delta\ell) = \Phi_\tau(\Omega) \cdot \mathcal{D}_{\text{loc}} = \Phi_\tau(\Omega) \exp\left(-\frac{2\Gamma_{\text{tot}}\Delta\ell}{cN_{\text{eff}}\rho_\tau}\right). \tag{4}$$

Theorem 1 (Mimicry of Exponential Gravitational Decoherence)**.** *In the weak-to-moderate baseline regime ($\Delta\ell \lesssim \mathcal{L}_{\text{sim}}$), relativistic synchronization lag induces a linear fractional visibility deficit $\Delta V_{\text{sync}}/\Phi_\tau \approx \Delta\ell/\mathcal{L}_{\text{sim}}$, which mimics first-order spacetime-foam or continuous spontaneous localization (CSL) decoherence rates $\Gamma_{\text{QG}} \propto \Delta\ell$.*

Proof. Expanding $V_{\text{null}}(\Omega, \Delta\ell, \Gamma_{\text{tot}})$ to first order in $\Delta\ell/\mathcal{L}_{\text{sim}} \ll 1$ yields:

$$V_{\text{null}} = \Phi_\tau(\Omega) \left[1 - \frac{2\Gamma_{\text{tot}}\Delta\ell}{cN_{\text{eff}}\rho_\tau} + \mathcal{O}\left(\frac{\Delta\ell^2}{\mathcal{L}_{\text{sim}}^2}\right) \right]. \tag{5}$$

If an experimental analysis omits the round-trip signaling factor $2\Delta\ell/(cN_{\text{eff}})$ by assuming naive point-locality ($\mathcal{D}_{\text{loc}} = 1$), the residual visibility drop $\delta V = \Phi_\tau(\Omega) - V_{\text{obs}}$ scales linearly with baseline $\Delta\ell$ and laser linewidth Γ_{tot}, producing an apparent anomalous decoherence rate $\Gamma_{\text{apparent}} = \Gamma_{\text{tot}}\Delta\tau_{\text{sig}}/\tau_{\text{int}}$ directly proportional to baseline separation. □

3 The Two-Tier Quantum Gravity Anomaly Monitor (Ξ_{QG})

To rigorously separate genuine quantum-gravity decoherence from relativistic synchronization lag, we define the **Two-Tier Quantum Gravity Anomaly Monitor** (Ξ_{QG}), implemented directly in Panel 6 of the `aeon-t` relativistic metrology simulation suite.

Definition 2 (Two-Tier Ξ_{QG} Anomaly Observable)**.** *Given an experimentally measured differential fringe visibility $V_{\text{obs}}(\Omega, \Delta\ell, \Gamma_{\text{tot}})$ and the calibrated null-hypothesis baseline $V_{\text{null}}(\Omega, \Delta\ell, \Gamma_{\text{tot}})$, the normalized Quantum Gravity Anomaly Monitor Ξ_{QG} is defined as:*

$$\Xi_{\text{QG}} \equiv \ln\left(\frac{V_{\text{null}}(\Omega, \Delta\ell, \Gamma_{\text{tot}})}{V_{\text{obs}}(\Omega, \Delta\ell, \Gamma_{\text{tot}})}\right) = -\ln\left(\frac{V_{\text{obs}}}{\Phi_\tau(\Omega)}\right) - \frac{2\Gamma_{\text{tot}}\Delta\ell}{cN_{\text{eff}}\rho_\tau}. \tag{6}$$

When V_{obs} is governed solely by classical signaling latency and technical phase noise, $V_{\text{obs}} = V_{\text{null}}$ and $\Xi_{\text{QG}} \equiv 0$ identically across all baselines and noise levels. Any statistically significant positive deviation ($\Xi_{\text{QG}} > 0$) signals excess non-classical decoherence beyond the Operational Simultaneity Horizon bound.

3.1 Two-Tier Diagnostic Architecture

Table 1: Two-Tier Ξ_{QG} Diagnostic Classification Protocol

Diagnostic Tier	Ξ_{QG} **Threshold**	**Noise-Sweep Behavior** ($\partial\Xi_{\text{QG}}/\partial\Gamma_{\text{tot}}$)	**Physical Interpreta**
Null Regime	$\Xi_{\text{QG}} \approx 0$ ($< 3\sigma_{\text{stat}}$)	$\partial\Xi_{\text{QG}}/\partial\Gamma_{\text{tot}} = 0$	Pure relativistic signa
Tier 1 Alert	$3\sigma_{\text{stat}} \leq \Xi_{\text{QG}} < 5\sigma_{\text{stat}}$	Non-zero residual under Γ_{tot} sweep	Unmodeled classical/t
Tier 2 Anomaly	$\Xi_{\text{QG}} \geq 5\sigma_{\text{stat}}$	$\partial\Xi_{\text{QG}}/\partial\Gamma_{\text{tot}} = 0$ (invariant)	Candidate spacetime

Discrimination Criterion 1 (Orthogonal Noise-Sweep Invariance Test)**.** *A candidate Tier 2 quantum-gravity anomaly is falsified as a classical synchronization artifact unless Ξ_{QG} remains invariant under controlled modulation of the technical laser phase-noise rate Γ_{tot}:*

$$\left.\frac{\partial\Xi_{\text{QG}}}{\partial\Gamma_{\text{tot}}}\right|_{\Delta\ell,\Omega} = 0. \tag{7}$$

4 Application to MAGIS-100, AION, and MIGA

For a 100 m vertical vacuum gradiometer ($N_{\text{eff}} = 1$, $\Delta\ell = 100$ m), the two-way light transit time is $\Delta\tau_{\text{sig}} \approx 667$ ns. While negligible for low-frequency mechanical drifts, modern optical clock transitions (^{87}Sr at 429 THz) driven by ultra-narrow lasers with effective high-frequency phase-noise rates $\Gamma_{\text{tot}} \sim 10^3 - 10^4$ s^{-1} and tight phase-locking thresholds ($\rho_\tau = 0.1$) exhibit an Operational Simultaneity Horizon of $\mathcal{L}_{\text{sim}} \approx 1.5$ km to 15 km.

At $\Delta\ell = 100$ m with $\mathcal{L}_{\text{sim}} = 1.5$ km, $\mathcal{D}_{\text{loc}} = \exp(-100/1500) \approx 0.9355$, producing an immediate 6.45% deterministic visibility reduction ($V_{\text{null}} = 0.9355\,\Phi_\tau$) purely from relativistic signaling lag. Scaling to a 1 km terrestrial shaft (AION-km / MAGIS-km) increases this baseline reduction to $\mathcal{D}_{\text{loc}} = \exp(-1000/1500) \approx 0.5134$ (48.7% visibility loss). Evaluating Ξ_{QG} subtracts this exact exponential envelope prior to statistical hypothesis testing.

5 Philosophical Implications for Spacetime Metrology

Within a neutral four-dimensionalist ontology, spacetime possesses a fixed metric structure $g_{\mu\nu}$ without privileged global present slices. Testing whether quantum mechanics modifies this classical spacetime manifold requires strict epistemic hygiene: observers cannot treat their own synchronization channels as infinitely rigid, zero-latency rods. By embedding $\mathcal{D}_{\text{loc}}$ into the null hypothesis via Ξ_{QG}, relativistic metrology separates the finite thermodynamic cost of establishing operational simultaneity from genuine non-classical curvature or quantum-gravity fluctuations.

6 Conclusion

Long-baseline quantum interferometers offer unprecedented sensitivity to the boundary between quantum theory and gravitation, provided their classical relativistic baselines are calibrated without naive point-locality approximations. The Classical Synchronization Null-Hypothesis Visibility V_{null} and the Two-Tier Anomaly Monitor Ξ_{QG} supply an exact, falsifiable calibration standard for current and next-generation quantum gravity experiments.

AI Collaboration & Authorship Transparency Statement

This work is the result of human-AI scholarly collaboration. The core philosophical framework, conceptual architecture, operational definitions, and theoretical direction were conceived and authored by **Luís Pimenta Damásio** (ORCID: 0009-0001-7606-7557). Mathematical formalization, analytical verification, and structural drafting support were provided by **Gemini** (Google AI) under the direct supervision, review, and editorial authority of the human author.

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